



S1.2 The nuclear atom

Structure 1 · Models of the particulate nature of matter · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Atoms consist of a tiny, dense, positively charged nucleus (containing protons and neutrons) surrounded by negatively charged electrons occupying most of the atom's volume. The atomic number (Z) equals the number of protons and defines the element; the mass number (A) is the sum of protons and neutrons. Isotopes are atoms of the same element with different numbers of neutrons (and therefore different mass numbers) but identical chemical properties, since chemistry is governed by electron number/arrangement. Mass spectrometry ionises, accelerates, deflects and detects atoms/molecules according to their mass-to-charge ratio, allowing isotopic abundances and relative atomic masses to be determined.

Worked example: follow the reasoning

Naturally occurring boron consists of two isotopes: ^{10}B (mass 10.01, abundance 19.9%) and ^{11}B (mass 11.01, abundance 80.1%).

Step 1: Relative atomic mass = $\Sigma(\text{isotope mass} \times \text{fractional abundance})$.

Step 2: $A_r = (10.01 \times 0.199) + (11.01 \times 0.801)$.

Step 3: $A_r = 1.992 + 8.819 = 10.81$.

Answer: $A_r(\text{B}) = 10.81$, consistent with the periodic table value.

Quick check · multiple choice

An atom has 17 protons, 18 neutrons and 17 electrons. What is this species?

A A chloride ion, $^{35}\text{Cl}^-$

B A neutral chlorine atom, ^{35}Cl

C A neutral chlorine atom, ^{37}Cl

D A sulfide ion, $^{35}\text{S}^{2-}$

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 Magnesium has three naturally occurring isotopes: ^{24}Mg , ^{25}Mg and ^{26}Mg . [6]
- (a) Define the term isotope. [2]
- (b) A sample of magnesium is analysed in a mass spectrometer and gives the following relative abundances: ^{24}Mg 79.0%, ^{25}Mg 10.0%, ^{26}Mg 11.0%. Calculate the relative atomic mass of this magnesium sample to 3 significant figures. [2]
- (c) Outline why isotopes of the same element have identical chemical properties despite differing in mass. [2]

Final answer: _____



Answers and explanation

Question 1 [6]

(a) Isotopes are atoms of the same element (same number of protons/atomic number) that have different numbers of neutrons, hence different mass numbers. [2]

(b) $\text{Ar} = (24 \times 0.790) + (25 \times 0.100) + (26 \times 0.110) = 18.96 + 2.50 + 2.86 = 24.3$ (3 s.f.). [2: correct method, correct final value]

(c) Chemical properties depend on the number and arrangement of electrons (particularly valence electrons), which is the same for all isotopes of an element since they all have the same number of protons/electrons; only the number of neutrons (and hence mass) differs. [2]

Quick check answer

B. Protons = 17 identifies chlorine ($Z = 17$); mass number = $17 + 18 = 35$; equal protons and electrons ($17 = 17$) means the atom is neutral.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Chemistry HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.